# Dong Yoon Lee

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## RESEARCH INTERESTS

Internet of Things (IoT); Vibration and acoustic signal processing; Multimodal sensing systems and signal processing; Pervasive and autonomous sensing; Embedded systems; Machine learning for physical and human-centered sensing

## EDUCATION

**Ph.D. in Electrical Engineering and Computer Science** Fall 2023–Present
*University of California, Merced — Merced, CA*

**B.S. in Computer Science and Engineering** 2019–2022
*University of California, Irvine — Irvine, CA*

## HONORS, AWARDS, AND PROFESSIONAL SERVICE

2025 Best Paper Award, ACM International Workshop on Intelligent Acoustic Systems and Applications (IASA '25)
2024 Artifact Evaluation Committee Member, ACM SenSys '24
2024 Web Chair, ACM BuildSys '24
2024 Artifact Evaluation Committee Member, ACM MobiSys '24
2024 Travel Grant Recipient, ACM HotMobile '24
2023 Best Poster Award, ACM SenSys '23
— Participant, UC Irvine ACM algorithm programming competition
— Participant, UC Irvine Hackathon; project: "Corona Watch"
2018 Third Place, Fresno State Mathematics Competition
2016 Tenth Place, Fresno State Integration Bee
— Regional First Place and Small School Third Place, Academic Decathlon

## PEER-REVIEWED PUBLICATIONS

12. **D. Y. Lee**, T. Patil, S. Pan, and A. Weakley. "Only Speaking FACTs: Foundation Models for Activity-Centric Timeline Description." *Proceedings of the 4th International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys '26)*, pp. 119–124, 2026.
https://doi.org/10.1145/3842436.3843810

**Lead contributions and technology**
- Introduced the **Activity-Centric Timeline (ACT) Description** task, which turns privacy-preserving IoT and structural-vibration streams into temporally grounded, natural-language accounts of daily routines for nonexpert remote-care users.
- Designed the hierarchical FACT pipeline: cross-modal signal-distribution alignment, foundation-model construction of structured activity representations, and instruction-guided LLM description generation.
- Used publicly available near-surface audio to reduce dependence on labeled vibration data, then evaluated the system in two real residential deployments.
- Achieved semantic textual similarity above 0.861 and ROUGE-1 F1 scores of 0.710 and 0.627, retaining useful timeline descriptions even when upstream activity predictions were imperfect.

11. **D. Y. Lee**, A. M. Weakley, T. Patil, H. Wei, D. Cardona, and S. Pan. "AlzGood: Validating Remote Care Systems in the Wild via an Interactive Deployment Framework." *2026 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, pp. 511–516, 2026.
https://doi.org/10.1109/CHASE69719.2026.00087

**Lead contributions and technology**
- Designed an LLM-driven interactive framework for deploying and validating remote-care vibration systems in lived-in homes, guided by the geriatric **4Ms** framework.
- Built **Interactive Installation**: it parses sensor specifications, home descriptions, and user preferences into a hierarchical graph, then uses chain-of-thought prompting to generate and score placements for usability, signal quality, and privacy.
- Built **Interactive Validation**: event detection, signal-similarity matching, pseudo-labeling, dialogue history, and context-aware speech prompts combine to collect scalable in-the-wild labels.
- Demonstrated feasibility in the home of an 88-year-old person with dementia; the LLM-selected deployment kept 80% of sensors untampered and produced labeling accuracy comparable to a sensor-context baseline while receiving the highest usability rating.

10. H. Wei, **D. Y. Lee**, S. Rohal, Z. Hu, R. Rossi, S. Fang, and S. Pan. "A Survey of Foundation Models for IoT: Taxonomy and Criteria-Based Analysis." *CCF Transactions on Pervasive Computing and Interaction*, pp. 1–29, 2025.
https://doi.org/10.1007/s42486-025-00208-w

9. **D. Y. Lee**, A. Weakley, H. Wei, B. Brown, K. Carrion, and S. Pan. "RRAR: Robust Real-World Activity Recognition with Vibration by Scavenging Near-Surface Audio Online." *Proceedings of the 3rd ACM International Workshop on Intelligent Acoustic Systems and Applications (IASA '25)*, pp. 1–6, 2025. **(Best Paper Award)**
https://doi.org/10.1145/3737901.3768365

**Lead contributions and technology**
- Curated task- and modality-aligned near-surface ASMR audio from the Internet to pretrain activity recognition when labeled structural-vibration data are scarce.
- Designed a multitask variational autoencoder with a CNN encoder, a temporal-convolutional activity classifier operating on the mean latent representation, and STFT reconstruction from sampled latent variables to separate activity content from user and environment variation.
- Developed the **ASMR–Vibration Adapter**, a compact low-rank 1D CNN, and a task-selective fine-tuning strategy that updates the classifier while freezing most of the pretrained model.
- Evaluated on real-world participants with 71%, 75%, 67%, and 50% accuracy; the method averaged 72% and improved over comparison methods by 13% on average.

8. S. Rohal, J. Zhang, F. Montgomery-Yale, **D. Y. Lee**, S. Schertz, and S. Pan. "Self-Adaptive Structure Enabled Energy-Efficient PM2.5 Sensing." *Proceedings of the 13th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems*, pp. 15–20, 2025.
https://doi.org/10.1145/3722572.3727928

7. S. Du, H. Wei, **D. Y. Lee**, Z. Hu, and S. Pan. "Graph-Based Physics-Guided Urban PM2.5 Air Quality Imputation with Constrained Monitoring Data." *ACM Transactions on Sensor Networks*, 2025.
https://doi.org/10.1145/3734869

**Supporting technical evidence**
- Developed a graph-based, physics-guided approach to fine-grained urban PM2.5 imputation under sparse monitoring coverage, with emphasis on California's disadvantaged San Joaquin Valley.
- Encoded physical knowledge through graph structure, edge features, and model layers so spatial relationships remain useful when observations are constrained.
- Improved MSE, MAE, and $R^2$ by 9–56% over comparison methods across the paper's evaluation settings.

6. S. Rohal, **D. Y. Lee**, P. Nguyen, and S. Pan. "GEM: Gear-based Environment-Integrated Mobility for Adaptive Indoor Human Sensing." *UbiSense: Emerging Techniques in Ubiquitous Sensing and Interaction*, 6 pages, 2026.
https://arxiv.org/abs/2505.10546

**Selected technical contributions**
- Created a hybrid gear matrix embedded in the environment so infrastructure-mounted sensors can move and adapt without requiring users to wear devices or introducing a free-roaming robot.
- Designed, formally analyzed, and implemented constrained path-finding algorithms, including modified A-star search and Transfer/Swap operations with parity and collision constraints.
- Validated the mobility concept with a 3×3 hardware prototype and simulations scaling to 64×64 matrices with multiple concurrent sensors.

5. S. Rohal, **D. Y. Lee**, C. Ruiz, J. Zhang, J. Fagert, J. Han, and S. Pan. "Don't Crosstalk to Me: Origami Structure-Augmented Sensing for Scalable Surface Pressure Monitoring." *Proceedings of the 22nd ACM Conference on Embedded Networked Sensor Systems (SenSys '24)*, November 4–7, 2024, Hangzhou, China, 14 pages.
https://doi.org/10.1145/3666025.3699353

**Selected technical contributions**
- Combined a Miura-Ori three-dimensional metasurface, which physically isolates sensing units and suppresses crosstalk, with multiplexed readout requiring only $H + W$ electrodes instead of $H \times W$.
- Designed, implemented, and evaluated the circuit-guided convolutional neural network (CG-CNN) that encodes the multiplexing topology to remove ghost activations.
- Achieved up to 2× the sensor-activation detection F1 score of comparison methods while retaining scalable surface-pressure sensing.

4. **D. Y. Lee**, Y. Li, Y. Zhang, H.-C. Wang, A. M. Weakley, and S. Pan. "KAP: Kinetic Augmented Pill Bottle for Vibration-Based Medication Interaction Recognition." *Proceedings of the 25th International Workshop on Mobile Computing Systems and Applications (HotMobile '24)*, February 28–March 1, 2024, San Diego, CA, USA, 6 pages.
https://doi.org/10.1145/3638550.3641140

**Lead contributions and technology**
- Proposed embedding medication identity directly into physical kinetic signatures using a battery-free, 3D-printed **Kinetic Augmentation Attachment** with a spring and bouncing mass for standard pharmacy pill bottles.
- Designed and implemented **KAFE**: STFT analysis, 700–800 Hz band-energy aggregation, and threshold-based peak, interval, and decay feature extraction followed by an RBF-kernel SVM.

- Evaluated multiple attachment configurations across five placement locations, reaching up to 99% supervised accuracy and 82% average leave-one-location-out accuracy—up to 27% above ablations and baselines.
- Preserved standardized pill bottles and existing ambient surface-vibration sensing without batteries, networking, or changes to the patient's medication-taking workflow.

**3.** Y. Zhang, S. Patel, **D. Y. Lee**, P. Celli, A. Bonde, and S. Pan. "Poster Abstract: LEVO: LEGO® Bricks Enhanced Single-Point Vibration Sensing for Occupant Monitoring." *Proceedings of the 21st ACM Conference on Embedded Networked Sensor Systems (SenSys '23)*, November 12–17, 2023, Istanbul, Türkiye, 2 pages. **(Best Poster Award)**
https://doi.org/10.1145/3625687.3628385

**Selected technical contributions**
- Used configurable LEGO-based metamaterial arrays to encode source-direction information into vibration modes observable at a single sensing point.
- Conducted experiments and signal analysis that established structure-assisted vibration acquisition as a practical direction for occupant monitoring.

**2.** **D. Y. Lee**, Z. Hu, Q. Do, P. Nguyen, and S. Pan. "Demo Abstract: Real-Time Teeth Functional Occlusion Monitoring via In-Mouth Vibration Sensing." *2022 21st ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*, pp. 505–506, 2022.
https://ieeexplore.ieee.org/document/9825953

**Lead contributions and technology**
- Implemented a real-time, end-to-end demonstration of teeth-contact vibration capture through retainer-form in-mouth sensors, with live signal processing and visualization for functional-occlusion monitoring.
- Integrated compact acquisition, event detection, and feedback components to translate the TeethVib sensing concept into an interactive IPSN demonstration.

**1.** S. Pan, **D. Y. Lee**, J. H. Lee, and P. Nguyen. "TeethVib: Monitoring Teeth Functional Occlusion Through Retainer Vibration Sensing." *2021 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, pp. 92–96, 2021.
https://ieeexplore.ieee.org/document/9697959

**Supporting technical evidence**
- Built a retainer sensing platform using four 1 cm × 1 cm, 125 μm piezoelectric-film elements sandwiched between two custom 0.6 mm retainers.
- Used Gaussian energy anomaly detection and relative energy profiles across channels to identify and interpret teeth-contact events.
- Evaluated natural and near-optimal functional occlusion across four dental models, producing vibration signatures designed to be interpretable by dental professionals.

## ADDITIONAL PUBLICATIONS

**A1.** H. Wei, **D. Y. Lee**, and S. Pan. "FROST: Foundation-Model Representations with Domain Knowledge for Spatially and Temporally Robust Frost Prediction." *Proceedings of ACM Sustainability Week Companion '26*, pp. 443–447, 2026.
https://doi.org/10.1145/3765611.3815487

**Supporting technical evidence**
- Combined zero-shot time-series foundation-model representations with direct temperature forecasts and hand-engineered meteorological features in a lightweight downstream predictor.
- Evaluated multi-year California weather-station data across 3–24 hour forecast horizons, targeting both spatial generalization to unseen stations and temporal robustness across seasons.
- Improved frost-prediction accuracy and calibration over conventional forecasting baselines while retaining a compact model suitable for agricultural decision support.

## PREPRINTS AND MANUSCRIPTS

**P2.** **D. Y. Lee**, A. Weakley, H. Wei, D. Cardona, and S. Pan. "Home Health System Deployment Experience for Geriatric Care Remote Monitoring." *arXiv preprint arXiv:2601.17608*, 2026. **(Earlier manuscript related to the peer-reviewed AlzGood paper [11].)**
https://arxiv.org/abs/2601.17608

**Lead contributions and technology**
- Engineered a plug-and-play wall-powered sensing node around an ESP32-S3, ADS131M02 ADC, and geophone, sampling approximately 6.8–7 kHz to capture 10–1000 Hz structural vibration.
- Built a deployment networking stack with low-overhead UDP transport, packet-level parity recovery, WireGuard tunneling, and a local edge hub; field deployments processed roughly 70–150 GB per day.
- Transferred a temporal-convolutional activity model from online acoustic data to vibration and developed a deployment-quality protocol organized around the geriatric **4Ms**.
- Used a graph representation and chain-of-thought LLM reasoning to balance SNR, privacy, usability, and tamper resistance over three deployment iterations in homes that included people with dementia.
- Reduced sampling-frequency standard deviation to 316 Hz from 734 and 799 Hz in earlier iterations while showing that LLM-recommended placements could balance signal quality and resistance to disturbance.

**P1.** H. Wei, **D. Y. Lee**, S. Rohal, Z. Hu, R. Rossi, S. Fang, and S. Pan. "A Survey of Foundation Models for IoT: Taxonomy and Criteria-Based Analysis." *arXiv preprint arXiv:2506.12263*, 2025. **(Superseded by the journal article listed above.)**
https://arxiv.org/abs/2506.12263

## PATENTS

**PT1.** S. Pan, Y. Zhang, **D. Y. Lee**, Y. Li, A. M. Weakley, and H.-C. Wang. "System and Method for Vibration-Based Medication Interaction Recognition." U.S. Patent Application 18/941,135, 2025.
https://patents.google.com/patent/US20250152091A1/en

## RESEARCH EXPERIENCE

**Structure-Assistive Design for More Effective Sensor Signal Acquisition** Fall 2023–Present

Effectively acquiring high-quality sensor signals that contain useful information for inference tasks is essential for IoT applications. However, sensor signals from real-world applications are often noisy and have limited capability for extracting comprehensive information. For example, structural vibration sensing used for unobtrusive in-home monitoring faces significant interference from the signal propagation channel. This work explores physical structures that manipulate signal generation and propagation to improve sensing performance.

- Leveraging metamaterials to embed directional information into vibration signals that propagate through a surface before reaching the sensor.
  *Role: Conducted experiments and signal analysis.*
- Leveraging kinetic augmentation attachments to embed medication identity into pill-bottle and ambient-surface interaction-induced vibration signals.
  *Role: Designed and implemented the kinetic-aware time-frequency feature extraction (KAFE) algorithm.*
- Creating a gear-based assistive structure to enable sensor mobility for wearable systems.
  *Role: Designed, proved, and implemented constrained path-finding algorithms for the gear-based structure.*

**Physical Knowledge-Guided Deep Learning for More Accurate Modeling with Limited Data** Spring 2024–Summer 2024

Deep learning is widely used for IoT applications, but training a robust model often requires a significant amount of labeled data. This research explores deep-learning architectures that incorporate physical knowledge to improve learning efficiency.

- Developed a circuit-guided convolutional neural network (CG-CNN) to mitigate the "ghosting" effect in multiplexed sensor matrices.
  *Role: Designed, implemented, and evaluated the CG-CNN architecture.*

**Integrating Signal Processing into Tokenization for IoT Signal-Based Question Answering** Summer 2024–Present

One challenge in the adoption of IoT technology is the lack of efficient interfaces between IoT sensing systems and users. Natural language provides an intuitive interaction modality. This project investigates ambient-information question answering over IoT data by integrating signal-processing techniques into efficient tokenization and learned data representations.

- Developing AI methods that integrate signal-processing techniques to extract meaningful data representations while minimizing computational requirements.
  *Role: Lead system design, implementation, and evaluation.*

## PUBLIC TALKS AND DEMONSTRATIONS

Kinetic Augmented Pill Bottle for Vibration-Based Medication Interaction Recognition 2024

*ACM HotMobile '24*

Real-Time Teeth Functional Occlusion Monitoring via In-Mouth Vibration Sensing 2022

*ACM/IEEE IPSN Demo*

GraPhy: Graph-Based Physics-Guided Urban Air Quality Modeling for Monitoring-Constrained Regions 2024

*ACM BuildSys '24*

## SKILLS AND KNOWLEDGE

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|---|---|
| **Foundation Models and LLMs** | Time-series and audio foundation models, cross-modal alignment, instruction-guided generation, chain-of-thought deployment support, representation transfer |
| **Machine Learning** | Supervised and unsupervised learning, CNNs, TCNs, variational autoencoders, graph learning, SVMs, regularization, domain adaptation |
| **Sensing Systems** | Structural vibration, acoustic, pressure, air-quality, and wearable sensing; multimodal sensor fusion; in-home field deployments |
| **Embedded Hardware** | Circuit design, ESP32-class sensor nodes, ADC and geophone acquisition, piezoelectric sensors, embedded software, data acquisition |
| **Physical Computing** | Kinetic mechanisms, origami structures, metamaterials, gear matrices, structure-assisted sensing, constrained path finding |
| **Programming Languages** | Python, C, C++, Java, JavaScript, Lisp, Prolog, Rust |
| **Signal Processing** | Vibration and acoustic analysis, STFT and time-frequency methods, event detection, feature extraction, signal-similarity analysis |
| **Networking and Edge Systems** | UDP sensor streaming, packet-level recovery, WireGuard tunneling, local edge processing, high-volume IoT data pipelines |
| **Operating Systems** | Resource allocation, synchronization, concurrent programming, Linux |